Journal of Geophysical Research

Supporting Information for

Revisiting Seismicity Criticality: A New Framework for Bias Correction of Statistical Seismology Model Calibrations

Jiawei Li[1], Didier Sornette[1], Zhongliang Wu[1,2], Jiancang Zhuang[1,3], and Changsheng Jiang[4]

[1]Institute of Risk Analysis, Prediction and Management (Risks-X), Academy for Advanced Interdisciplinary Studies, Southern University of Science and Technology (SUSTech), Shenzhen, China.

[2]Institute of Earthquake Forecasting, China Earthquake Administration, Beijing, China.

[3]The Institute of Statistical Mathematics, Research Organization of Information and Systems, Tokyo, Japan.

[4]Institute of Geophysics, China Earthquake Administration, Beijing, China.

Contents of this file

Introduction

In the present study, we identify and quantify three sources of bias: (i) boundary effects, (ii) finite-size effects, and (iii) censorship, which cause errors in seismic analysis and predictions. By employing a variant of the ETAS model with variable spatial background rates, we propose a method to correct for these biases, focusing on the branching ratio n, a key indicator of earthquake triggering potential. We validate our method using synthetic earthquake catalogs, accurately recovering the true branching ratio n_{true} after correcting biases with n_{app}. Applying our framework to the earthquake catalogs of California, New Zealand, the China Seismic Experimental Site (CSES), and Noto Peninsula in Japan, we revisit seismicity's criticality to enhance our comprehension of seismic patterns, aftershock predictability, and inform earthquake risk mitigation and management strategies. The Supporting Information accompanying this study provides an in-depth statistical characterization of seismicity in the four study regions, along with the calibrated parameters of the Epidemic-Type Aftershock Sequences (ETAS) model used to simulate synthetic catalogs.

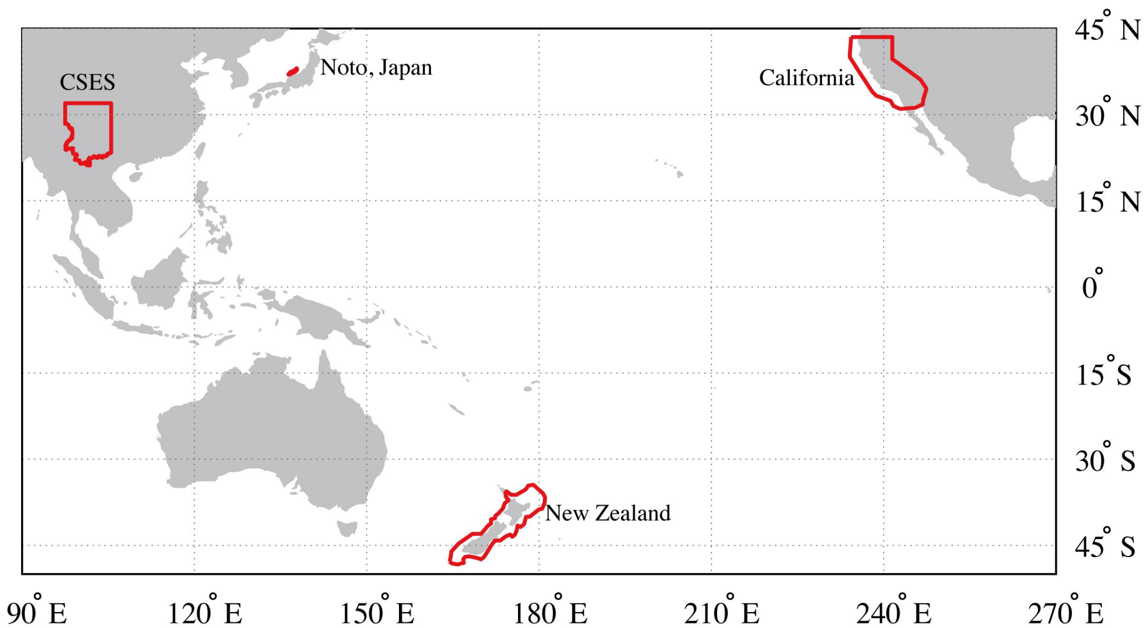

Figure S1. Map showing the spatial extents of the four study regions analyzed in this study: California, New Zealand, the China Seismic Experimental Site (CSES), and Noto of Japan. Basic information for each region—including area, data source, primary period, and seismicity size—is provided in Table S1. Statistical characterizations of seismicity in each region are shown in Figures S2–S4.

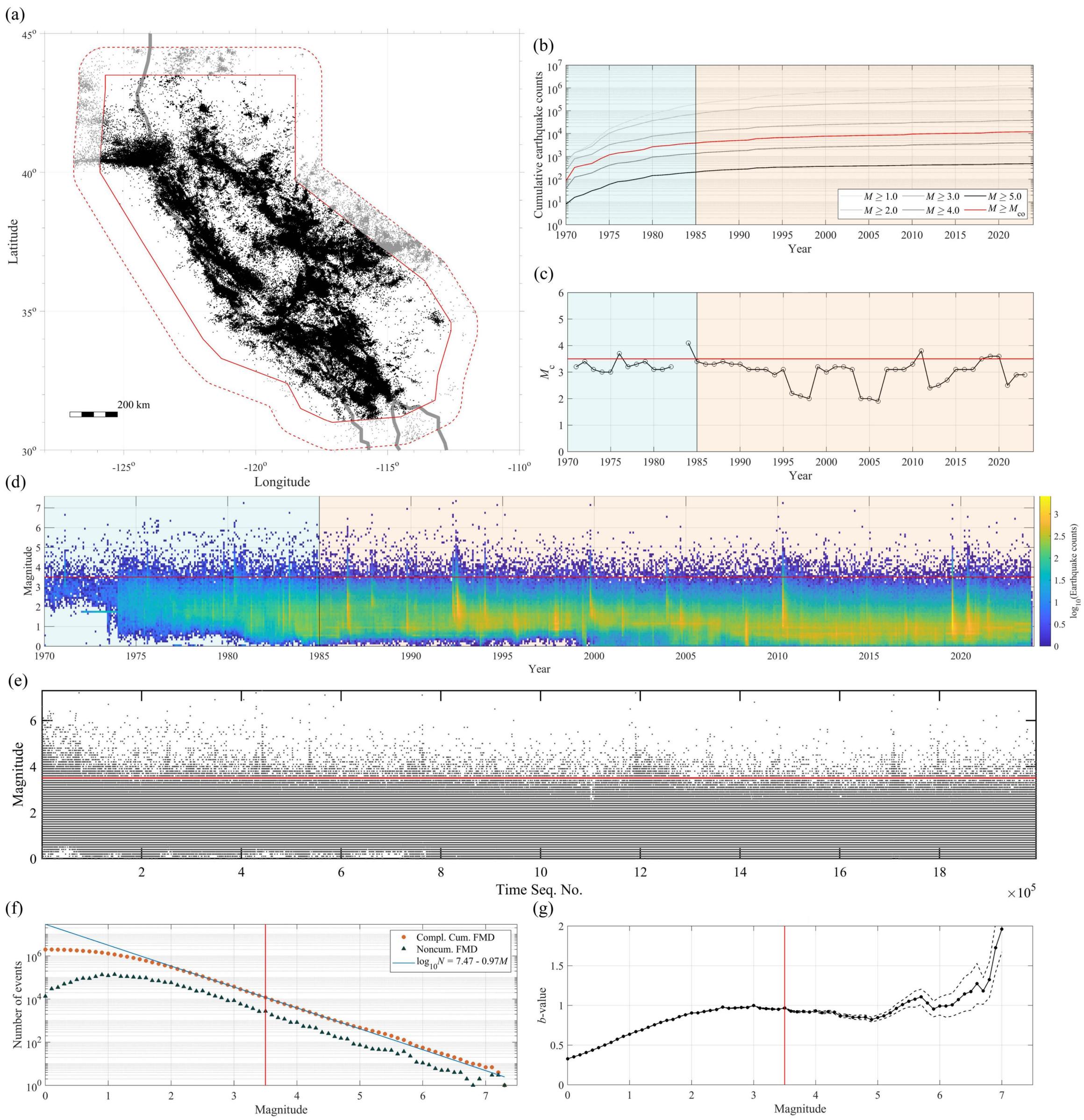

Figure S2. (a) Epicenter distribution of recorded earthquakes from 1970 to 2023 in the primary California region (solid red line), complemented by an approximate 100 km auxiliary space band (dashed red line). The thick gray line delineates the coastline. (b) Time series depicting cumulative earthquake counts for different cut-off magnitudes ($M_{co} \geq 1, \geq 2, \geq 3, \geq 4$, and ≥ 5). (c) Evolution of the completeness magnitude (M_c) over time. (d) Monthly earthquake counts per 0.1 magnitude bin. (e) Magnitude–time sequence plot. (f) Complementary cumulative and density frequency–magnitude distribution alongside the Gutenberg–Richter (GR) law fitted using earthquakes with magnitudes larger than M_{co}^{best} = 3.5. (g) Variation of the *b*–

value with M_{co}, with the thin dashed line indicating the ± 1σ standard deviation. The red lines in panels (b) to (g) denote M_{co}^{best} = 3.5 in California. The light red and light blue shaded regions in panels (b) to (d) represent the primary period and the auxiliary time band, respectively.

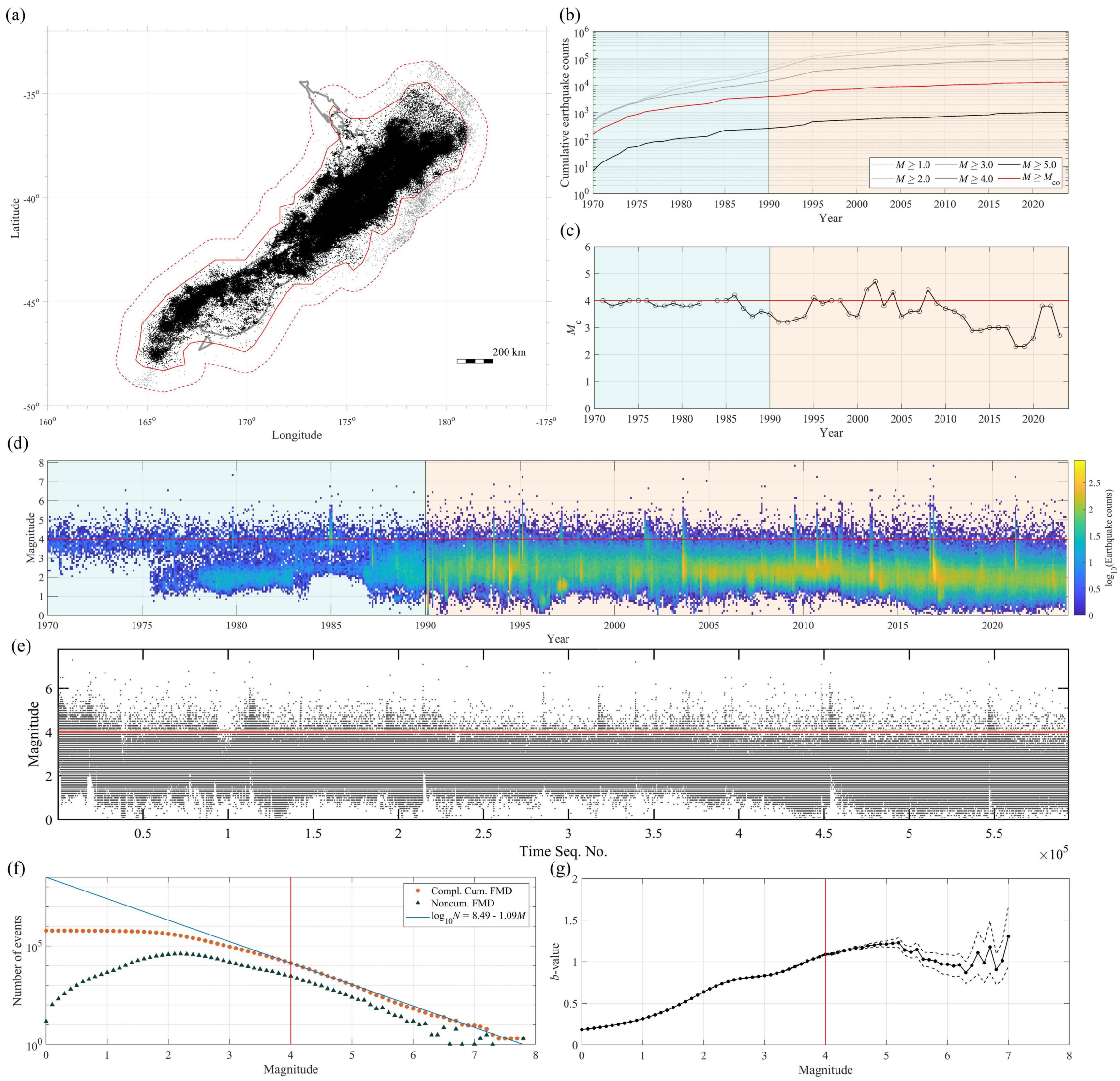

Figure S3. (a) Epicenter distribution of recorded earthquakes from 1970 to 2023 in the primary New Zealand region (solid red line), complemented by an auxiliary space band (dashed red line). Panels (b) to (g) are the same as in Figure S2, with the red lines denoting M_{co}^{best} = 4 in New Zealand.

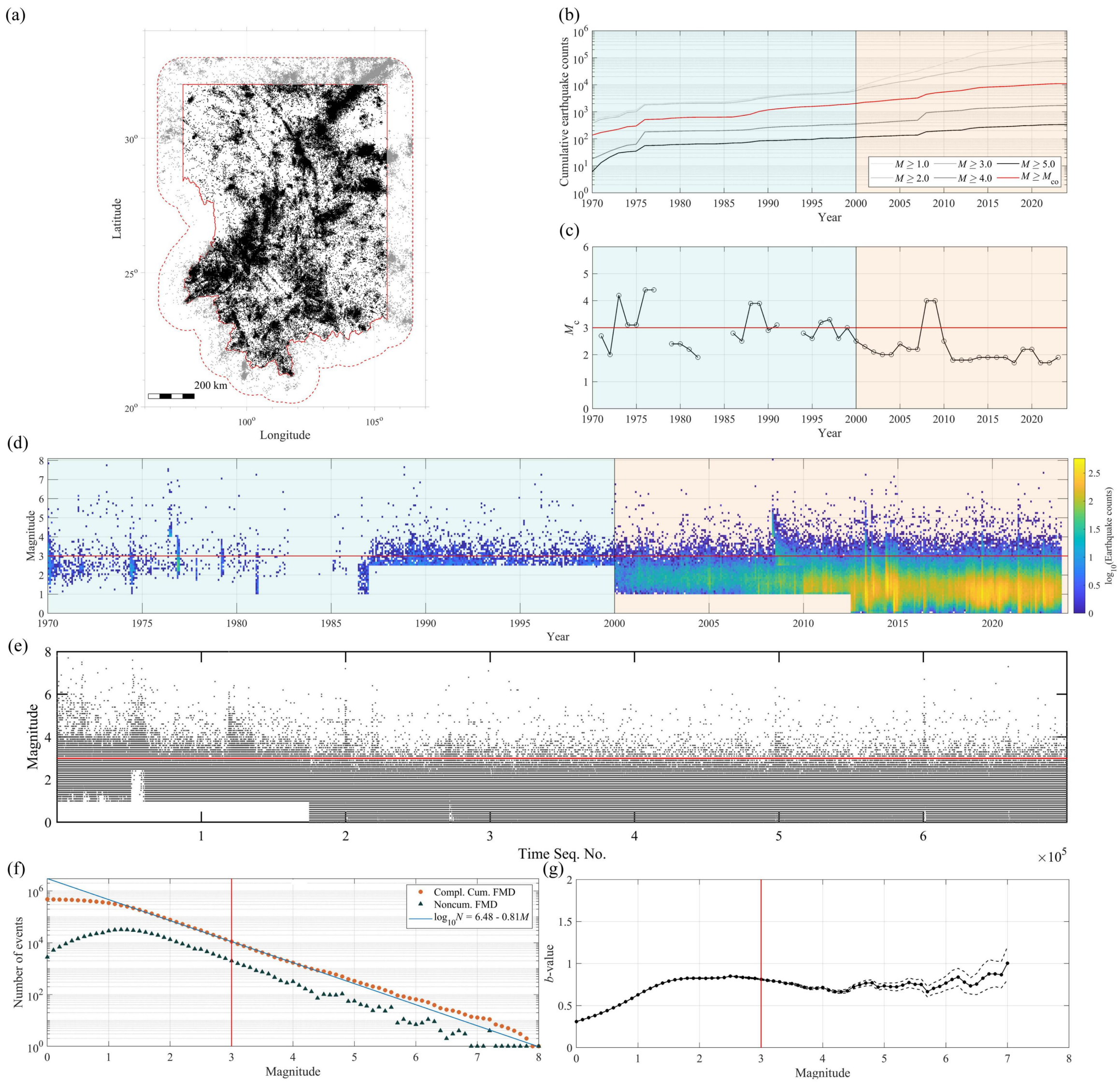

Figure S4. (a) Epicenter distribution of recorded earthquakes from 1970 to 2023 in the primary region of the China Seismic Experimental Site (CSES; solid red line), complemented by an auxiliary space band (dashed red line). Panels (b) to (g) are the same as in Figures S2 and S3, with the red lines denoting M_{co}^{best} = 3 in the Sichuan–Yunnan region, China.

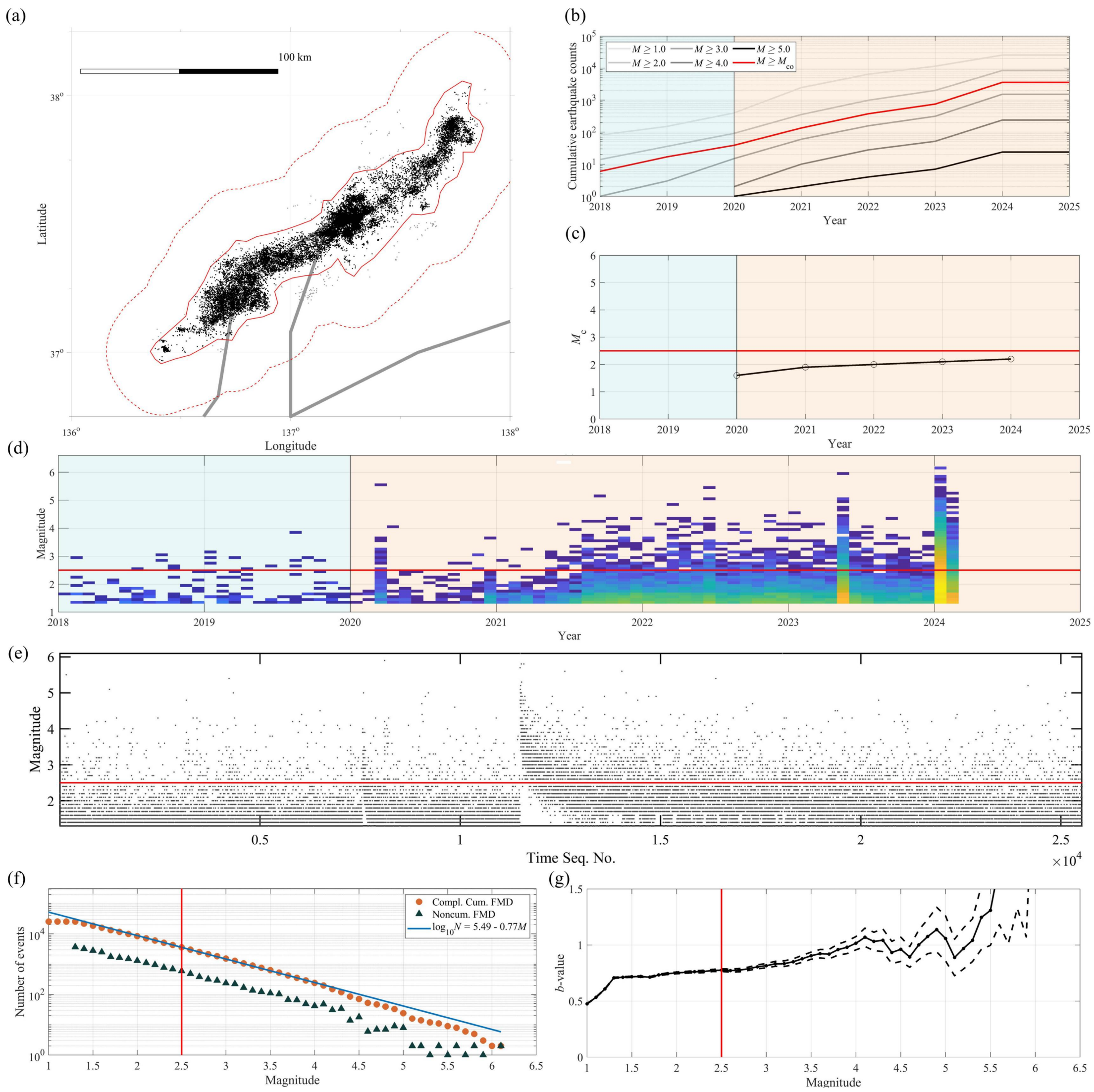

Figure S5. (a) Epicenter distribution of relocated earthquakes from 2018 to 2024 in the primary region of the Noto Peninsula in Japan (solid red line), complemented by an auxiliary space band (dashed red line). Panels (b) to (g) are the same as in Figures S2, S3 and S4, with the red lines denoting M_{co}^{best} = 2.5 in the Noto Peninsula, Japan.

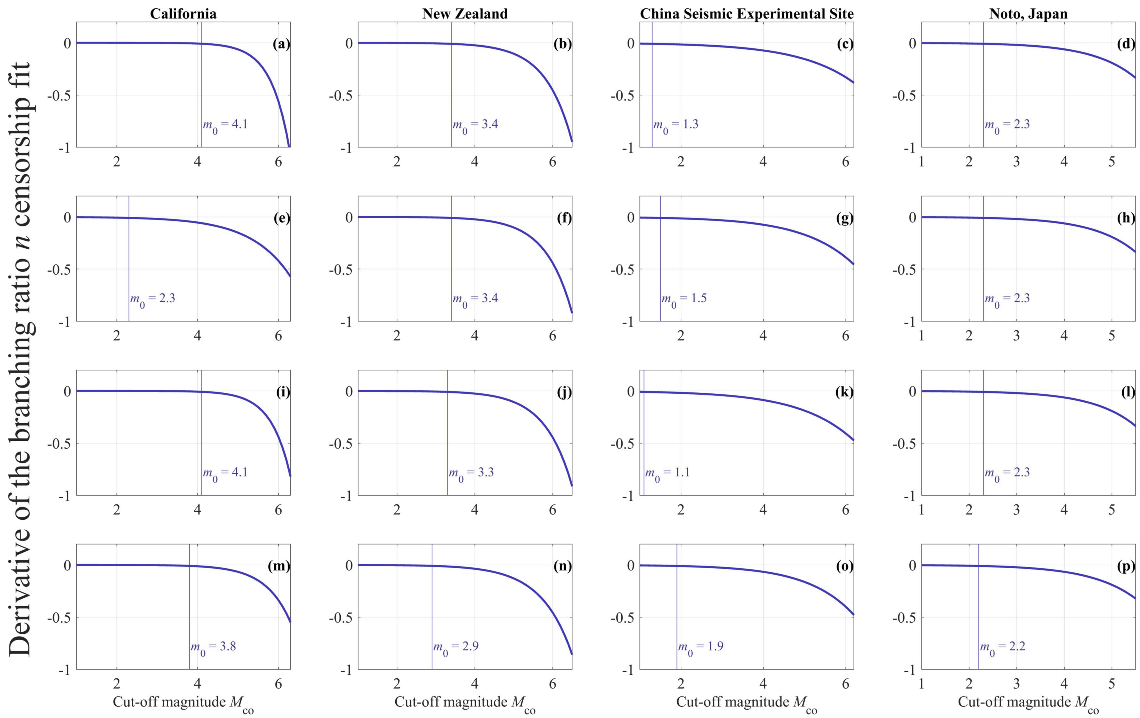

Figure S6. Derivatives of the branching ratio curves $n_{app}(M_{co})$ shown in Figure 4 of the model used to correct the censorship effect as a function of M_{co}. From left to right, the columns represent California, New Zealand, CSES, and Noto of Japan. The panels, ordered from top to bottom, represent: (1) The primary catalog with an auxiliary spatio–temporal band; (2) The primary catalog with an auxiliary spatial band only; (3) The primary catalog with an auxiliary temporal band only; (4) The primary catalog combined with the catalog from the auxiliary region, with an auxiliary temporal band. The model is represented by Formula (15), and the fitted parameter values are listed in Table 1. The values of M_{co} at which the derivatives become smaller than −0.01 (greater than 0.01 in absolute value) are identified as m_0, as explained in Section 5.2 of the main text.

Table S1. Main characteristics of the primary catalog used in this study.

	Area [km^2]	Source	M_{co}^{best}	Primary period	Primary duration [year]	$N_{evt}^{primary}$
California	961,240	ANSS	3.5	1985/01/01 – 2023/11/02	38.80	8,265
New Zealand	694,460	GeoNet	4.0	1990/01/01 – 2023/12/12	33.94	9,762
CSES	781,240	CENC	3.0	2000/01/01 – 2023/08/23	23.63	9,163
Noto, Japan	3,997	Peng et al. (2024)	2.5	2020/01/01 – 2024/02/19	4.14	3,584

M_{co}^{best}: the best or smallest magnitude of completeness above which earthquake data are considered complete; CSES: the China Seismic Experimental Site. The areas extend beyond the political borders. For more details about the catalog source, please refer to the Data and Resources section.

Table S2. ETAS model parameters calibrated form observed catalogs at M_{co}^{best} and utilized for simulating seismicity in the study regions.

	Scenario	N_{bkg}	Φ	$\log_{10}K$	α	$\log_{10}d$	$1+\rho$	γ	$\log_{10}c$	$1+\omega$	$\log_{10}\tau$	$\log_{10}D$	Q	b
California	**1**	2,517.73	−6.73	−0.41	1.03	−0.39	0.69	1.34	−2.70	1.04	3.84	1.20	0.64	1.00
	2	2,894.03	−6.67	−0.50	1.11	−0.39	0.70	1.35	−2.69	1.04	3.20	1.20	0.68	1.00
	3	2,423.85	−6.75	−0.39	1.11	−0.40	0.70	1.36	−2.71	1.03	3.88	1.20	0.62	1.00
	4	2,772.60	−6.86	−0.39	1.05	−0.42	0.64	1.40	−2.69	1.03	3.90	1.20	0.54	0.97
New Zealand	**1**	3324.34	−6.41	−0.51	1.42	1.18	0.97	0.69	−2.21	1.14	3.58	2.02	1.03	1.08
	2	3430.56	−6.40	−0.56	1.45	1.18	0.97	0.69	−2.21	1.14	3.28	2.00	1.00	1.08
	3	3258.05	−6.42	−0.48	1.44	1.18	0.98	0.68	−2.22	1.14	3.63	2.04	1.03	1.08
	4	3530.52	−6.62	−0.47	1.40	1.23	0.92	0.63	−2.24	1.13	3.61	1.97	0.83	1.07
CSES	**1**	3012.10	−6.35	−0.54	0.95	1.20	2.48	0.54	−3.01	0.88	2.83	1.70	0.79	0.84
	2	3049.79	−6.34	−0.57	1.00	1.20	2.49	0.54	−3.03	0.87	2.79	1.69	0.79	0.84
	3	2588.77	−6.42	−0.38	0.91	1.19	2.31	0.53	−3.00	0.89	3.00	1.90	0.87	0.84
	4	2966.04	−6.56	−0.37	0.90	1.19	2.03	0.49	−3.05	0.86	3.03	1.81	0.59	0.84
Noto, Japan	**1**	187.18	−4.49	−0.04	0.78	−0.19	1.01	0.87	−2.08	0.95	1.42	0.60	9.99	0.78
	2	187.19	−4.49	−0.41	1.19	−0.18	1.01	0.87	−2.08	0.95	1.42	0.60	9.99	0.78
	3	186.33	−4.49	−0.03	0.77	−0.19	1.01	0.87	−2.08	0.95	1.42	0.60	9.99	0.78
	4	180.90	−5.02	−0.41	1.20	−0.23	0.97	0.91	−2.14	0.93	1.37	0.60	3.88	0.78

Note: Parameters c and τ are given in days; Φ is the background rate per km^2 per year on a logarithmic scale.